

ASSESSMENT TOOL 2

CONCEPT MAPPING

1. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks: a. Show relationships between these concepts b. Analyze how each contributes to secure communication c. Include one real-world application for each.

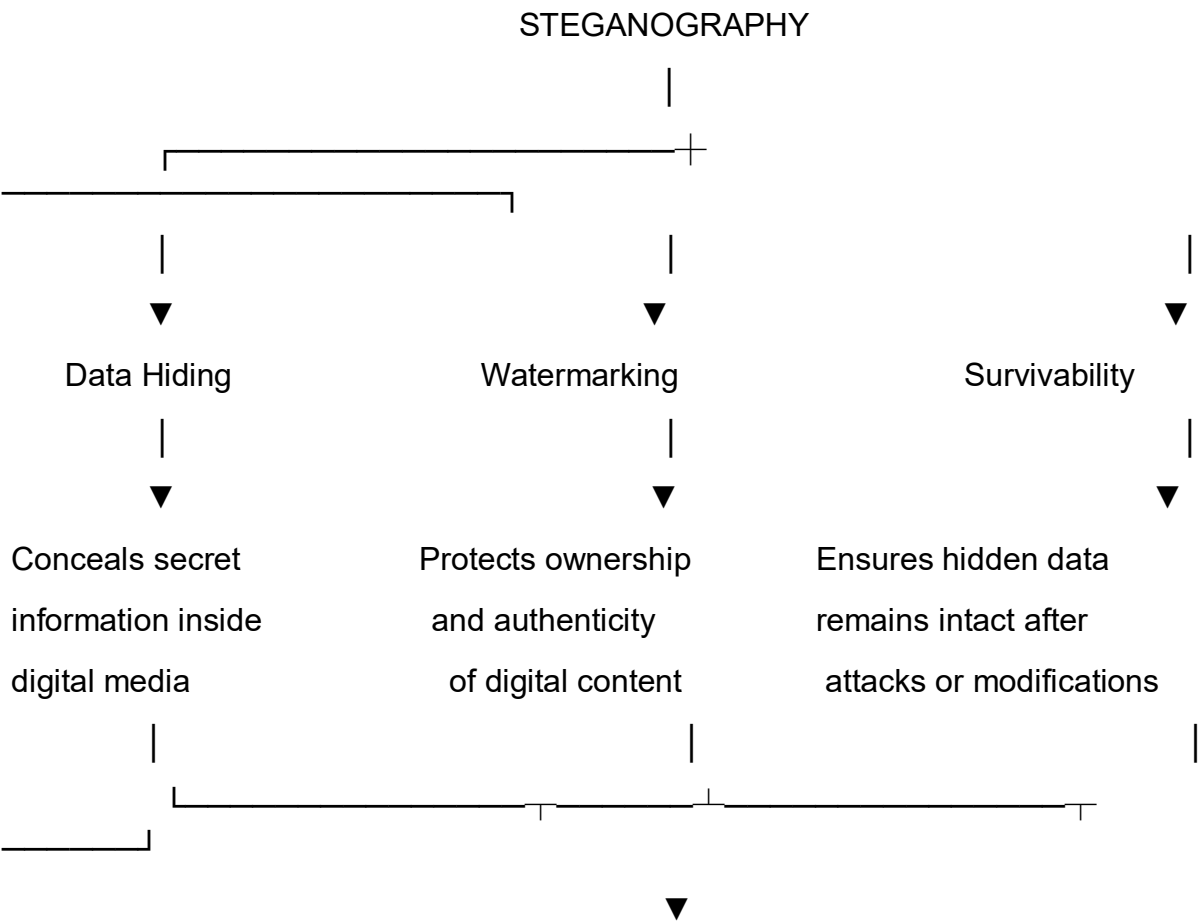
Concept Map: Steganography Concepts

Aim :

To understand the relationship between **Data Hiding, Watermarking, and Survivability** in steganography and analyze their roles in secure communication.

Theory :

Steganography is the practice of concealing secret information within digital media such as images, audio, video, or text so that the existence of the message remains hidden. It is widely used for secure communication, copyright protection, and authentication.



Secure Communication

a. Relationships Between the Concepts

- **Steganography** is the main technique used to hide information.
 - **Data Hiding** is the process of embedding secret information into a cover medium.
 - **Watermarking** is a special form of data hiding used to prove ownership and authenticity of digital content.
 - **Survivability** refers to the ability of the hidden information or watermark to remain intact even after compression, editing, or attacks.
 - Together, these concepts improve the confidentiality, authenticity, and reliability of digital communication.
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b. Contribution to Secure Communication

Concept	Contribution to Secure Communication
Data Hiding	Keeps confidential information invisible to unauthorized users.
Watermarking	Protects digital content from copyright infringement and verifies ownership.
Survivability	Ensures hidden information remains recoverable even after modifications or attacks.

c. Real-World Applications

Concept	Real-World Application
Data Hiding	Sending confidential military or government messages inside digital images.
Watermarking	Copyright protection for photographs, videos, movies, and digital documents.
Survivability	Secure transmission of medical images where hidden patient information remains intact after image compression.

Result :

The concepts **Data Hiding**, **Watermarking**, and **Survivability** are interconnected components of steganography. Data Hiding conceals information, Watermarking protects ownership and authenticity, and Survivability ensures the hidden information

remains intact even after processing or attacks. Together, they enhance secure communication in various real-world applications.

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks: a. Show how each concept is mathematically connected b. Map their role in encryption/decryption c. Analyze which concept is most critical for public key cryptography and justify

Concept Map: Number Theory Concepts Used in Cryptography

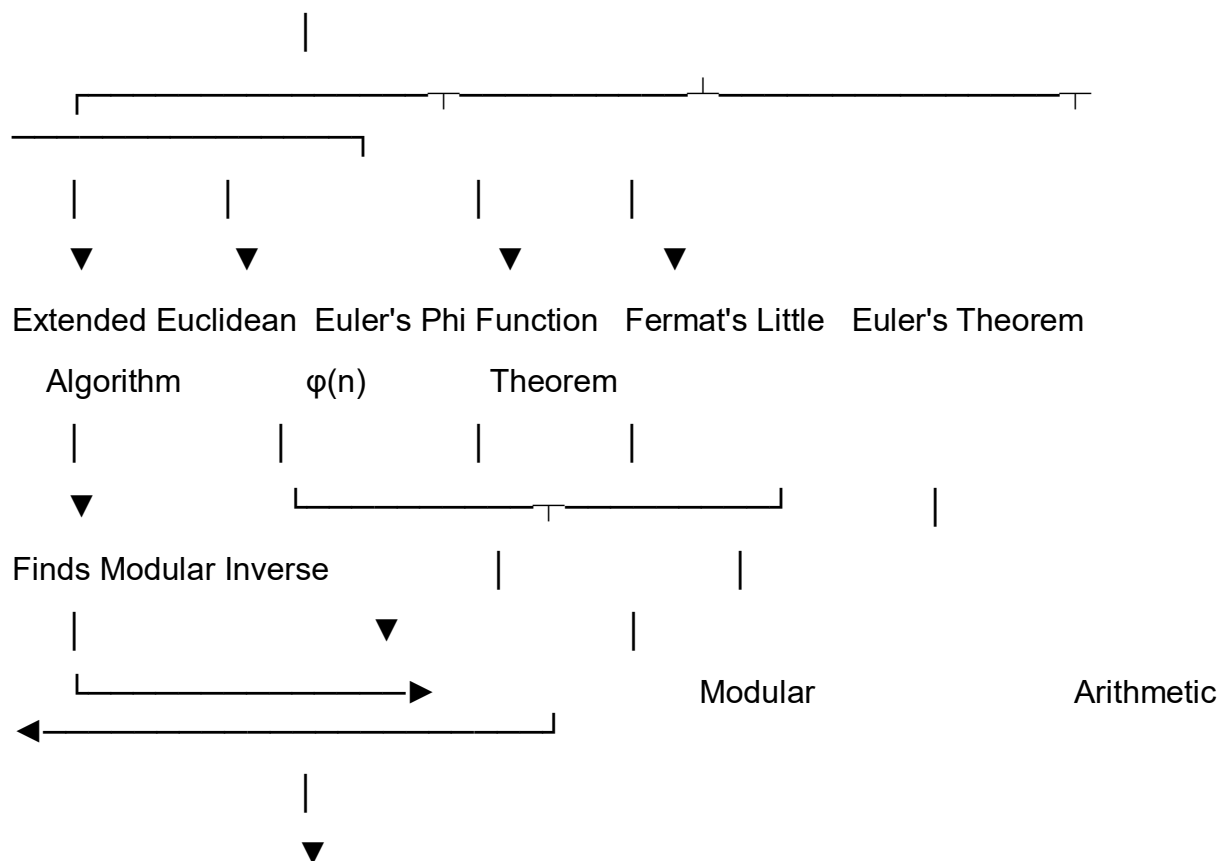
Aim :

To understand the relationships among **Modular Inverse**, **Extended Euclidean Algorithm**, **Fermat's Little Theorem**, **Euler's Phi Function**, and **Euler's Theorem**, and analyze their importance in public key cryptography.

Theory :

Number theory forms the mathematical foundation of modern cryptography. Concepts such as the **Extended Euclidean Algorithm**, **Modular Inverse**, **Euler's Phi Function**, **Euler's Theorem**, and **Fermat's Little Theorem** are widely used in encryption and decryption algorithms, especially in the RSA cryptosystem.

NUMBER THEORY



RSA Encryption & Decryption

a. Mathematical Connections

- The **Extended Euclidean Algorithm** is used to compute the **Modular Inverse** of a number.
- The **Modular Inverse** is required for finding the **private key** in RSA.
- **Euler's Phi Function ($\phi(n)$)** calculates the number of integers that are relatively prime to n .
- **Euler's Theorem** uses $\phi(n)$ and states that:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\gcd(a, n) = 1$.

- **Fermat's Little Theorem** is a special case of Euler's Theorem when n is a prime number:

$$a^{p-1} \equiv 1 \pmod{p}$$

- Together, these concepts form the mathematical basis for public key cryptographic algorithms such as RSA.

b. Role in Encryption and Decryption

Concept	Role in Cryptography
Extended Euclidean Algorithm	Computes the modular inverse used to generate the RSA private key.
Modular Inverse	Used to calculate the private decryption key from the public key.
Euler's Phi Function ($\phi(n)$)	Determines the number used in RSA key generation.
Euler's Theorem	Ensures correct encryption and decryption through modular arithmetic.
Fermat's Little Theorem	Simplifies modular exponentiation and supports efficient cryptographic computations.

c. Most Critical Concept for Public Key Cryptography

Among these concepts, the **Modular Inverse** is the most critical for public key cryptography.

Justification

- In the RSA algorithm, the private key **d** is calculated as the modular inverse of the public exponent **e** modulo $\phi(n)$.
- Mathematically,

$$d \equiv e^{-1} \pmod{\phi(n)}$$

- The **Extended Euclidean Algorithm** is used to compute this modular inverse efficiently.
 - Without the modular inverse, the private key cannot be generated, making decryption impossible.
 - Therefore, although all number theory concepts are important, the **Modular Inverse** is the key component that enables secure encryption and decryption in RSA.
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Result

The concepts **Extended Euclidean Algorithm**, **Modular Inverse**, **Euler's Phi Function**, **Euler's Theorem**, and **Fermat's Little Theorem** are mathematically interconnected and collectively provide the foundation for public key cryptography. Among them, the **Modular Inverse** is the most critical because it enables the generation of the RSA private key, making secure decryption possible.